

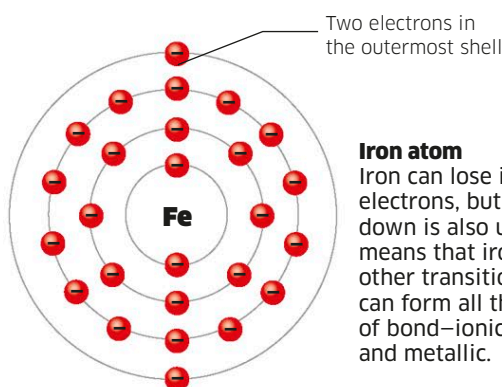
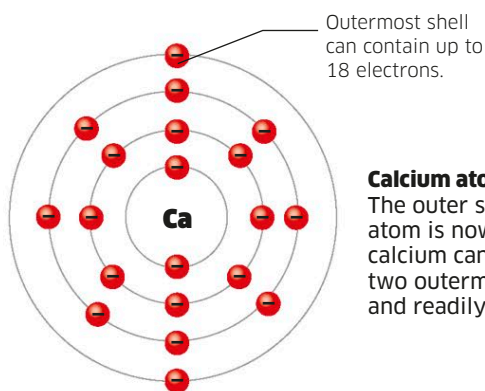
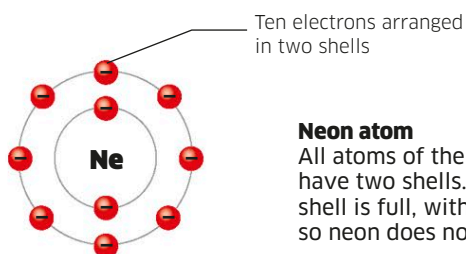
Bonding

Matter is made of atoms. Most of them are joined, or bonded, together. The bonds that hold atoms together are formed by the outermost parts of each atom: the electrons in the atom's outer shell.

There are three main types of bonding: ionic, covalent, and metallic. An ionic bond forms when electrons from one atom transfer to another, so that the atoms become electrically charged and stick together. A covalent bond forms when electrons are shared between two or more atoms. In a metal, the electrons are shared freely between many metal atoms. All chemical reactions involve bonds breaking and forming.

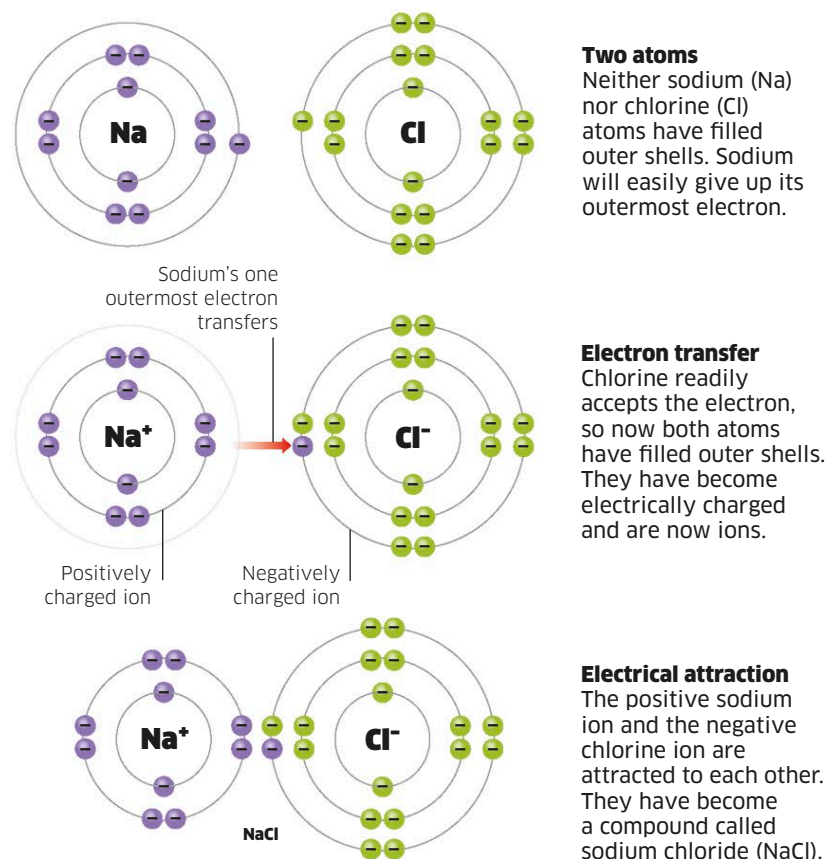
To bond or not to bond

The number of electrons an atom has depends upon how many protons are in its nucleus. This number is different for each element (see p.28). The electrons are arranged in "shells," and it is the electrons in the outermost shell that take part in bonding. An atom is stable when the outermost shell is full (see p.13). The atoms of some elements have outermost shells that are already full—they do not form bonds easily. But most atoms can easily lose or gain electrons, or share them with other atoms, to attain a full outer shell. These atoms do form bonds and take part in chemical reactions.

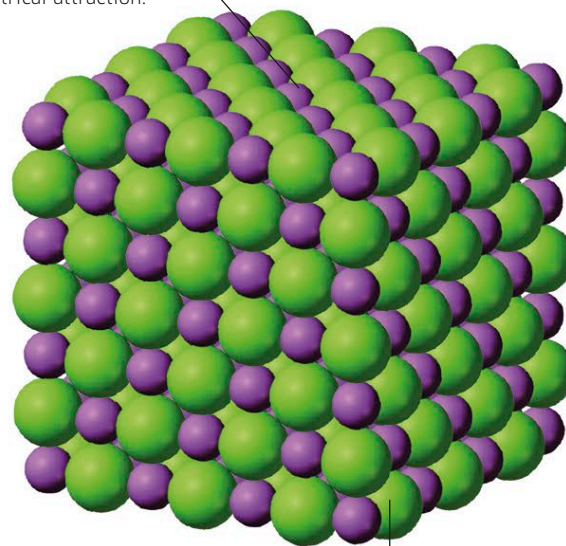


Ionic bonding

Many solids are made of ions: atoms, or groups of atoms, that carry an electric charge because they have either more or fewer negative electrons than positive protons. Ions form when atoms (or groups of atoms) lose or gain electrons in order to attain full outer electron shells. Electrical attraction between positive ions (+) and negative ions (-) causes the ions to stick together, forming a crystal.



Sodium ions and chlorine ions are held together by electrical attraction.



Ionic crystal

Ions of opposite electric charge are attracted to each other, and they form a regular pattern called a crystal. Many solids are ionic crystals, such as salt.

Salt crystal

The ions arrange in a regular pattern, forming a crystal of the compound sodium chloride (NaCl), or table salt.